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Air pollution, surrounding green, road proximity and Parkinson's disease: A prospective cohort study

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A R T I C L E I N F O

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Cohort study

A B S T R A C T

Background: Though growing evidence has linked air pollution to Parkinson's disease (PD), the results remain inconsistent. Less is known about the relevance of road proximity and surrounding green. We aimed to investigate the individual and joint associations of air pollution, road proximity and surrounding green with the incidence of PD in a prospective cohort study.
Methods: We used data from a prospective cohort of 47,516 participants recruited from July 2015 to January 2018 in Ningbo, China. Long-term exposure to particulate matter with aerodynamic diameter ≤2.5 μm ($PM_{2.5}$) and ≤10 μm (PM_{10}) and nitrogen dioxide (NO_2) estimated by land-use regression models, road proximity and surrounding green assessed by Normalized Difference Vegetation Index (NDVI) were calculated based on the residential address for each participant. Cox proportional hazard models were used to analyze the individual and joint effects of air pollution, road proximity, and surrounding green on PD.
Results: In single-exposure models, $PM_{2.5}$, PM_{10}, NO_2 and road proximity was associated with increased risk of PD (e.g. Hazard Ratio (HR) = 1.51, 95%CI:1.02, 2.24 per interquartile range (IQR) increase for $PM_{2.5}$) while surrounding green was associated with decreased risk of PD (e.g. HR = 0.80, 95%CI:0.65, 0.98 per IQR increase for NDVI in 300 m buffer). In two-exposure models, the associations of $PM_{2.5}$ and surrounding green persisted while the associations of NO_2 and road proximity attenuated towards unity.
Conclusions: We found that $PM_{2.5}$ were associated with increased risk of incident PD while surrounding green was associated with decreased risk of PD. Future studies about PD etiology may benefit from including multiple environmental exposures to address potential joint associations.

1. Introduction

Parkinson's disease (PD) is the second most common neurodegenerative disorder after Alzheimer's disease (Kalia and Lang, 2015). There are around 6 million people worldwide having PD and the prevalence has increased by 145% since 1990 estimated by the Global Burden of Disease (GBD) in 2016 (Collaborators, 2019). The prevalence and incidence of PD increased sharply with age and peaked after age 80 (Driver et al., 2009; Pringsheim et al., 2014). Given the increasing aging population worldwide, the disease burden of PD will have important public health implications. The etiology of PD remains unknown but is considered to be an interplay of both genetic and environmental factors (Kalia and Lang, 2015). Since there is no cure exist for PD, identification of modifiable risk factors is crucial for disease prevention.

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Long-term exposure to ambient air pollution has been linked to multiple health outcomes including mortality (Chen and Hoek, 2020), cardio-respiratory diseases (Cohen et al., 2017), and neurodegenerative diseases including PD (Fu et al., 2019). Toxicological studies showed that toxins in ambient air pollutants can promote brain inflammation and oxidative stress (Block et al., 2012; Segalowitz, 2008), which may contribute to the development of PD. Evidence also showed that air pollutants might exacerbate disease progression by accelerating these biological pathways or worsening intermediate processes (Gao et al., 2011). Despite there is experimental evidence, epidemiological studies showed inconsistent results on the associations between air pollution exposure and risk of PD. A systematic-review and meta-analysis including 13 studies up to 2018 reported pooled relative risks (RR) of PD incidence was 1.06 (95% confidence interval (CI):0.99, 1.14) and 1.01 (95%CI:0.98, 1.03) for per 10 μg/m^3 increase in long-term exposure to fine particulate matter ($PM_{2.5}$) and nitrogen dioxide (NO_2) respectively, and there was high heterogeneity between the study-specific results (Kasdagli et al., 2019). More evidence from longitudinal studies is needed to add to these findings.

Road proximity and surrounding green are usually spatially correlated with air pollution (Dadvand et al., 2014; Hystad et al., 2014). Epidemiological studies showed that road proximity (living near major roads) is associated with increased risk of dementia (Chen et al., 2017) and cognitive impairment (Yao et al., 2020). Besides, surrounding green has been reported to be positively associated with a broad range of health-outcomes (Chen et al., 2020; Ji et al., 2019; Klompmaker et al., 2019b).

The majority of previous studies on the associations of air pollution, road proximity and surrounding green with PD have only evaluated the effect of single-exposure, ignoring potential confounding by the other two factors, leaving the evidence for the combined effect scarce. Given the ubiquitous exposure of air pollution, road proximity and surrounding green and a growing aging population worldwide, a better understanding of the potential role of these environmental exposures on PD would provide important insight to reduce the disease burden of PD. The current study aims to investigate the individual and combined relationships between long-term exposure to air pollution, surrounding green, road proximity and risk of PD in a Chinese prospective cohort.

2. Methods

2.1. Study population

We did a prospective cohort study in Yinzhou District of Ningbo, China, which is a major port city in Eastern China covering 1381 km^2. From July 2015 to January 2018, a total of 48,908 individuals who were residents in the selected nine towns in the Yinzhou district and aged over 18 years old were invited to the local hospital to undertake a face-to-face interview and health examination. After excluding 1392 participants with duplicated records and 18 participants under 18 years old, 47,516 participants remained. Baseline demographic and lifestyle information was collected via a structured questionnaire by well-trained medical staff. More detailed information on the baseline investigation was described in Supplemental Methods. Follow-up was conducted via record linkage between the baseline database and the Yinzhou regional health information system (HIS) through the Wonders Big Data Management Platform. This platform integrates regional medical insurance system, chronic disease surveillance system, resident death registration system, and hospital outpatient and inpatients information system covering all major health service institutions, and has been applied in previous epidemiological studies (Li et al., 2019; Lin et al., 2018; Sun et al., 2020; Tang et al., 2019; Yu et al., 2019). All participants gave written informed consent for inclusion in the cohort before the baseline questionnaire investigation. Institutional Review Board of the Zhejiang University School of Medicine provided ethical approval.

2.2. Case ascertainment

Electronic medical records were retrieved from the regional HIS. A unique anonymously person identifier was used for each participant in the cohort to link to medical records. The medical records including person identifier, date of diagnosis, diagnosis name and diagnosis code (International Classification Disease 10, ICD-10) were collected between January 1st, 2009 and May 31, 2020. Cases of PD were defined as participants having at least one record with ICD-10 code (G20, G21.11, G21.19 and G21.8). The date of the first diagnosis was considered as the incident date. PD cases that occurred before the date of baseline investigation were considered prevalent cases (n = 251) and were subsequently excluded.

2.3. Covariates

Demographic and lifestyle factors were obtained through a structured questionnaire at baseline. Body mass index (BMI) was calculated as weight in kilograms divided by squared height in meters. Current smoking was defined as at least one cigarette per day for more than one year or at least five packs in one month. Current alcohol consumption was defined as alcohol intake reaches 100 g per week. Tea consumption was defined as more than two times a week for more than two months. Physical activity (defined as running, biking, brisk walking and other aerobic activities at least 20 min) was derived from the questionnaire and divided into four categories including sedentary, less than one time a week, 1–4 times a week, and more than four times a week.

2.4. Air pollution exposure

Monthly concentrations of particulate matter with aerodynamic diameter ≤10 μm (PM_{10}), $PM_{2.5}$ and NO_2 at the residential addresses of the participants were predicted by land-use regression (LUR) models. Model details have been described elsewhere (Zhang et al., 2018). Briefly, air pollution monitoring and meteorological data in Zhejiang Province, China from 2013 January to 2017 December were derived from the Chinese Ecology and Environment Ministry with technical support from the Qingyue environmental protection information technology service center (http://data.epmap.org/). Potential land use predictors including population density, distance to the nearest road, altitude, landcover (percentage of landcover within different buffers) derived from Geographic Information System (GIS) and time-varying meteorological data were used in the generalized additive regression to predict air pollution concentrations. The model performances were evaluated by the 10-fold cross-validation with R^2 values 0.749 for the $PM_{2.5}$ model, 0.691 for the PM_{10} model and 0.652 for the NO_2 model. Subsequently, these LUR models were assigned to the residential addresses of the participants. Cumulative average air pollution concentrations from 2013 January (which is the earliest date of available air pollution monitor data) to the date of the baseline survey were calculated for each participant.

2.5. Road proximity

Road information was derived from the OpenStreetMap (OSM), which is a geographic information platform that has been widely used in developed countries (Arsanjani et al., 2015). Studies have shown that the OSM data for eastern China is complete and with high positional accuracy (Zhang et al., 2015; Zheng and Zheng, 2014). We classified the OSM road data into four categories: A1 (free-ways, such as motorways and trunk ways, usually with limited access), A2 (primary roads, major roads that link towns or main road within cities), A3 (tertiary roads, such as residential roads, which serve as an access to housing or within a community), A4 (residential roads, pedestrian walkways, and tracks) (Zhang et al., 2018). In the current study, proximity to the nearest A1 or A2 roads was calculated for each participant's residential address using

ArcGIS.

2.6. Surrounding green

Exposure to surrounding green was assessed by the Normalized Difference Vegetation Index (NDVI). In brief, NDVI is a satellite-image-based index that calculates the ratio of the difference between the near-infrared region and red visible reflectance to the sum of these two measures. NDVI values range from −1.0 to 1.0, with larger values indicating higher levels of the vegetative index. Negative NDVI values were set to zero (Klompmaker et al., 2018). In the current study, NDVI measurements were obtained from the Moderate Resolution Imaging Spectro-Radiometer (MODIS) in the National Aeronautics and Space Administration's Terra Satellite (McMorris et al., 2015; Thiering et al., 2016). We linked the coordinate of each participants' residential address to the NDVI imagery and calculated the annual surrounding green in buffers with radii of 300 m and 1000 m.

2.7. Statistical analysis

The results are presented as percentages for categorical variables and means ± standard deviation (SD) for normally distributed continuous variables or median (25th-75th centiles) for non-normally distributed continuous variables. Spearman correlations between different environmental exposures were calculated. We used Cox proportional hazard regression models to analyze the associations of environmental exposures with the incident of PD. Calendar time was used as the time scale and each participant has one observation in the current analysis. The proportional risk assumption was tested using the Schoenfeld residuals method (All P values > 0.05).

We first performed single-exposure models in three models defined *a priori*: Model 1 adjusted for sex, age; Model 2 further adjusted for education level (illiterate, primary school, middle school, high school and above) and family income (<5000, ≥5000&<10000, ≥10000&<30000, ≥30000&<50000, ≥50000); Model 3 further adjusted for smoking status (current, former or never), alcohol consumption (current, former or never), tea drinking (current, former or never), BMI (underweight, normal weight, overweight or obesity), and physical activity (inactive, low, moderate and high); For each exposure variable, we used natural splines with 3 degree of freedom to explore the potential non-linear exposure-response relationships. Goodness of fit of the spline models and linear models were compared using the likelihood ratio test. As exposure-response curves for air pollutants and road proximity showed slight to moderate deviation from linearity, we presented the association estimates for all exposure variables in quartile (except road proximity) and in per interquartile range (IQR) increment. Statistical significance of a linear trend was tested by including the median value of the quartile of each exposure in the regression model as a continuous variable.

We performed additional analyses stratified by sex (male vs female), age (<65 years or≥65 years) and smoking status (ever smoker vs never smoker) to explore any potential effect modifier. Potential interactions were tested by including a multiplicative interaction term in the regression models. Sensitivity analyses were conducted among the participants without a history of cardiovascular disease and excluding participants with a follow-up period of less than 1 year. We further repeated analysis among the non-movers who did not move preceding 10 years before the baseline investigation. To test if there is any random cluster in the study region, we further performed random-effect Cox models by assigning the town as random intercept in the Cox model. Using the random-effect model allowed us to control for unmeasured factors affecting PD risk that are common to participants within a spatial cluster but may vary from different clusters.

Two-exposure models were conducted to analyze whether a potential association of one of the environmental exposures with PD was confounded by other exposures. Tri-exposure models (air pollutant + road proximity + NDVI) were also conducted. Generalized variance inflation factors were used to quantify potential multicollinearity in two-exposure and tri-exposure models (Fox and Monette, 1992).

We also used the cumulative risk index (CRI) method (Crouse et al., 2015; Jerrett et al., 2013) to analyze which combination of exposures is needed to fully capture the joint impact of air pollution, road proximity and surrounding green. The CRI method has been previously used in the combined analysis of air pollutants, surround green and road traffic noise on the cardiometabolic disease (Jerrett et al., 2013; Klompmaker et al., 2019b). The joint hazard ratios represent the combined estimates for a 1-unit increase in air pollution and road proximity and a 1-unit decrease in surrounding green exposure. A more detailed calculation of the joint hazard ratio was attached in Supplemental methods. Two-sided P-values <0.05 were considered as statistically significant. All analyses were conducted in R (version 3.6.3).

Table 1
Baseline characteristic of the study participants.

Characteristic	
Age[years], mean ± SD	62.27 ± 11.09
Female, n (%)	27320 (58.3)
Education level, n (%)	
Illiterate	15584 (33.3)
Primary school	20148 (43.0)
Middle school	8548 (18.2)
High school and above	2559 (5.5)
Family income [RMB], n (%)	
<5000	460 (1.0)
≥5000&<10000	2013 (4.3)
≥10000&<30000	15735 (33.6)
≥30000&<50000	14933 (31.9)
≥50000	13698 (29.2)
Marital status, n (%)	
Unmarried/Never married	880 (1.9)
Married	40923 (87.4)
Divorced	300 (0.6)
Widow	4705 (10.0)
BMI category, n (%)	
Underweight	12503 (26.7)
Normal	23350 (49.9)
Overweight	7651 (16.1)
Obesity	3335 (7.1)
Smoking status, n (%)	
Current	7577 (16.2)
Former	1738 (3.7)
Never	37524 (80.1)
Alcohol consumption status, n (%)	
Current	7758 (16.1)
Former	1738 (3.7)
Never	37524 (80.1)
Tea consumption status, n (%)	
Current	4153 (8.8)
Former	205 (0.4)
Never	38512 (82.2)
Physical activity, n (%)	
Inactive	13315 (28.4)
Low	18441 (39.4)
Moderate	10859 (23.2)
High	424 (9.0)
Environmental exposure	
$PM_{2.5}$ [μg/m^3], median (25th-75th centiles)	38.23 (33.64–41.83)
PM_{10}[μg/m^3], median (25th-75th centiles)	60.87 (52.42–63.21)
NO_2 [μg/m^3], median (25th-75th centiles)	34.67 (28.80–40.50)
Road proximity [m], median (25th-75th centiles)	152.63 (50.32–432.73)
NDVI 300 m, median (25th-75th centiles)	0.38 (0.26–0.51)
NDVI 1000 m, median (25th-75th centiles)	0.38 (0.30–0.50)

Note: SD, standard deviation; BMI, body mass index; IQR, interquartile range; $PM_{2.5}$, PM with aerodynamic diameter ≤2.5 μm; PM_{10}, PM with aerodynamic diameter ≤10 μm; NO_2, nitrogen dioxide; NDVI, normalized difference vegetation index.

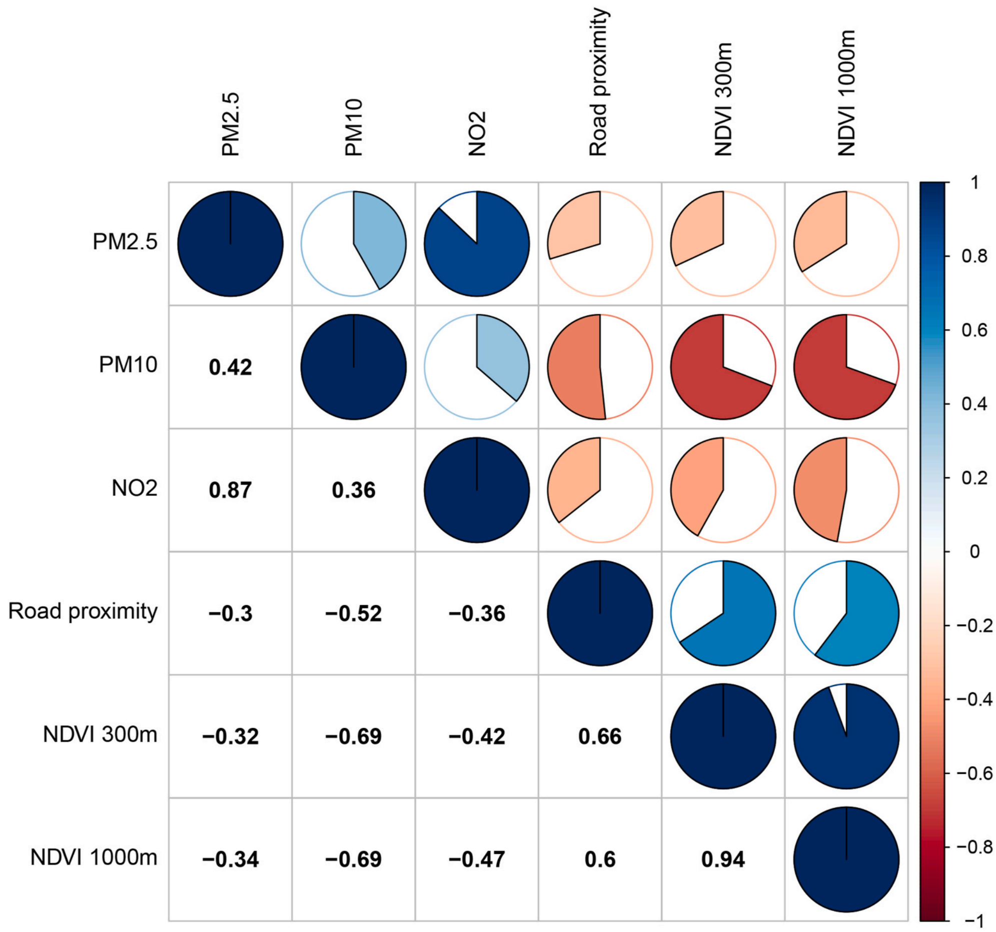

Fig. 1. Correlation (Spearman) matrix of $PM_{2.5}$, PM_{10}, NO_2, road proximity and surrounding green. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

3. Results

3.1. Basic characteristics

After excluding 426 participants without valid geocoded addresses and 251 prevalent PD cases, a total of 46,839 participants were included in the final analyses, of which 27,320 (58.3%) participants were female. The baseline age was 62.27 ± 11.09 years old. Most of the participants had a low level of education (illiterate or primary school) and were married. Half of the participants were in the normal range of BMI while 26.7% were underweight. Nearly 20% of the participants self-reported as current smokers or ever smokers. More than 60% of the participants reported an inactive or low level of physical activity (Table 1). During the median follow-up of 3.5 years and a total of 110,046 person-years, 206 incident PD cases were identified.

3.2. Exposure distribution and correlations

Median (25th-75th centiles) exposure level of air pollution was 38.23 (33.64–41.83), 60.87 (52.42–63.21), 34.67 (28.80–40.50) μg/m^3 for $PM_{2.5}$, PM_{10} and NO_2, respectively. The median distance to major roads was 152.63 (25th-75th centiles: 50.32–432.73) m. The median exposure of surrounding green in the buffer of 300 m and 1000 m were essentially the same (Table 1 and Table S1). In general, moderately negative correlations of residential air pollution concentrations with road proximity and surrounding green were observed (Fig. 1). Air pollutants were positively correlated ranging from 0.36 (PM_{10} with NO_2) to 0.87 ($PM_{2.5}$ with NO_2). Surrounding green in different buffer sizes were highly correlated.

3.3. Single-exposure regression models

We found positive associations of $PM_{2.5}$ and PM_{10} with the incident of PD, which was significant in the analysis with PM both as a continuous term (i.e. HR = 1.38, 95%CI:1.12, 1.85 per IQR increment for $PM_{2.5}$) and for the highest quartile compared with the lowest quartile (i.e. HR = 1.52, 95%CI: 1.03, 2.25 for $PM_{2.5}$; Table 2). A trend toward positive was also observed for the association between NO_2 and PD (HR = 1.22, 95%CI:0.95, 1.58 per IQR increment; Table 2).

We observed an inverse association of road proximity with PD incidence in the analysis of log-transformed distance as continuous terms or for the nearest two categories (<50 m or 50–100 m) compared with the reference category (>300 m) (i.e. HR = 1.59, 95%CI: 1.06, 2.37 for 50–100 m). We also observed that a higher level of surrounding green was associated with a lower risk of PD, with stronger associations observed for NDVI 300 m (HR = 0.80, 95%CI:0.68, 0.98 per IQR increment) compared with NDVI 1000 m (HR = 0.85, 95%CI:0.71, 1.01 per IQR increment) in terms of the effect size (Table 2).

Associations of air pollution, road proximity, and surrounding green with PD were robust by the adjustment for covariates, as the association estimates nearly remained unchanged when further adjusted for individual socioeconomic status and lifestyle factors (Table 2). The exposure-response curve for each environmental exposure was shown in Fig. 2. Generally, the shapes of the curve for air pollutants and log-transformed road proximity were flattened at the relatively low level but increase steadily at higher concentrations (decrease steadily for

Table 2
Adjusted associations of air pollution, road proximity and surrounding green with incident Parkinson's disease.

Exposure	Parkinson incidence rate[a]	Hazard Ratio (95%CI)		
		Model 1	Model 2	Model 3
$PM_{2.5}$, μg/m^3				
Q1 (≤33.64)	176.5	1.00 (Ref)	1.00 (Ref)	1.00 (Ref)
Q2 (>33.64 and ≤ 38.23)	139.1	0.87 (0.58, 1.31)	0.84 (0.56, 1.27)	0.84(0.55, 1.26)
Q3 (>.≫38.23 and ≤ 41.83)	196.3	1.16 (0.80, 1.69)	1.14 (0.78, 1.66)	1.14(0.78, 1.66)
Q4 (>41.83)	278.2	1.52(1.03, 2.25)	1.52(1.02, 2.25)	1.51(1.02, 2.24)
P for trend	–	0.06	0.05	0.05
Per IQR increment	–	1.38 (1.12, 1.85)	1.41 (1.14, 1.75)	1.41 (1.14, 1.75)
PM_{10}, μg/m^3				
Q1 (≤52.42)	180.5	1.00 (Ref)	1.00 (Ref)	1.00 (Ref)
Q2 (>52.42 and ≤ 60.87)	153.1	0.87(0.60, 1.27)	0.85(0.58, 1.24)	0.85(0.58, 1.24)
Q3 (>60.87 and ≤ 63.21)	195.9	1.04(0.71, 1.50)	1.01(0.69, 1.48)	1.01(0.68, 1.48)
Q4 (>63.21)	291.0	2.41(1.52, 3.82)	2.37(1.46, 3.85)	2.39 (1.47, 3.88)
P for trend	–	0.12	0.09	0.07
Per IQR increment	–	1.26 (1.01, 1.57)	1.25 (0.99, 1.56)	1.24 (0.99, 1.56)
NO_2, μg/m^3				
Q1 (≤28.80)	180.9	1.00 (Ref)	1.00 (Ref)	1.00 (Ref)
Q2 (>28.80 and ≤ 34.67)	155.2	0.81 (0.54, 1.20)	0.76(0.51, 1.14)	0.76(0.51, 1.14)
Q3 (>34.67 and ≤ 40.50)	186.8	1.04(0.71, 1.52)	0.97(0.66, 1.43)	0.97(0.66, 1.44)
Q4 (>40.50)	256.4	1.29(0.87, 1.93)	1.18(0.78, 1.78)	1.19(0.79, 1.79)
P for trend	–	0.41	0.39	0.38
Per IQR increment	–	1.22 (0.95, 1.58)	1.16 (0.88, 1.51)	1.16 (0.89, 1.51)
Road proximity, m				
<50	219.6	1.33 (0.89, 1.98)	1.32 (0.89, 1.98)	1.32 (0.88, 1.97)
50–100 m	246.8	1.59 (1.06, 2.37)	1.58 (1.05, 2.37)	1.57 (1.05, 2.37)
101–200 m	183.3	1.13 (0.74, 1.74)	1.15 (0.74, 1.78)	1.15 (0.74, 1.78)
201–300 m	125.2	0.76 (0.48, 1.22)	0.75 (0.47, 1.20)	0.75 (0.47, 1.20)
>300 m	180.3	1.00 (Ref)	1.00 (Ref)	1.00 (Ref)
P for trend	–	0.080	0.090	0.090
Log(Distance)	–	0.92 (0.83, 1.01)	0.92 (0.83, 1.01)	0.92 (0.83, 1.02)
NDVI 300 m				
Q1 (≤0.26)	249.5	1.00 (Ref)	1.00 (Ref)	1.00 (Ref)
Q2 (>0.26 and ≤ 0.38)	204.7	0.81 (0.55, 1.20)	0.80(0.54, 1.19)	0.80(0.54, 1.19)
Q3 (>0.38 and ≤ 0.51)	190.7	0.78(0.52, 1.18)	0.82(0.54, 1.23)	0.81(0.54, 1.23)
Q4 (>0.51)	142.6	0.54(0.36, 0.82)	0.56(0.37, 0.86)	0.57(0.37, 0.86)
P for trend	–	0.004	0.005	0.005
Per IQR increment	–	0.80 (0.68, 0.98)	0.80 (0.65, 0.98)	0.80 (0.65, 0.98)
NDVI 1000 m				
Q1 (≤0.30)	205.5	1.00 (Ref)	1.00 (Ref)	1.00 (Ref)
Q2 (>0.30 and ≤ 0.38)	227.5	1.10(0.74, 1.65)	1.13(0.75, 1.69)	1.12(0.74, 1.68)
Q3 (.≫0.38 and ≤ 0.50)	194.7	1.01(0.67, 1.50)	1.06 (0.71, 1.59)	1.06(0.70, 1.58)
Q4 (>0.50)	147.1	0.72(0.47, 1.08)	0.76(0.50, 1.17)	0.76(0.50, 1.17)
P for trend	–	0.076	0.168	0.178
Per IQR increment	–	0.85 (0.71, 1.01)	0.87 (0.72, 1.07)	0.87 (0.72, 1.07)

Note: IQR, interquartile range; PM2.5, PM with aerodynamic diameter ≤2.5 μm; PM10, PM with aerodynamic diameter ≤10 μm; NO2, nitrogen dioxide; NDVI, normalized difference vegetation index; Model 1 adjusted for age, sex; Model 2 further adjusted for marital status, education level and income; Model 3 further adjusted for BMI category, smoking, alcohol consumption, tea consumption and physical activity.

[a] Parkinson incidence rate was calculated as cases/person-years*100,000.

distance to roads). The exposure-response relationship between NDVI and PD was generally linear.

We observed similar directions of association between air pollution, road proximity, surrounding green, and incident of PD when analyses were stratified by sex (Table S2), by age (Table S3) or smoking status (Table S4). No significant interactions of the exposure and stratifying variables were found. Sensitivity analyses by excluding participants with CVD at baseline (N = 672), excluding participants with follow-up less than 1 year (N = 7726), repeated analysis among the non-movers 10 years preceding the baseline survey (N = 45,543) and assigning random-effects to the town of participants resided didn't materially change our findings in the main analyses (Tables S5–S8).

3.4. Multi-exposure regression models

As the surrounding green in 300 m and 1000 m were high correlated (r = 0.94) and the association estimates in single-exposure models were similar, we only reported NDVI 300 m in multi-exposure models. Associations of air pollution, road proximity and surrounding green with PD were slightly attenuated in the two-exposures models with larger confidence intervals compared with those in the single-exposure models (Fig. 3). The associations between NDVI 300 m and PD seemed to be stable when adjusted for air pollution or road proximity while the associations of NO_2 generally attenuated towards null when adjusted for other air pollutants, road proximity or surrounding green. Association estimates in tri-exposure models were largely unchanged compared with those in the two-exposure model (Table S9). As for CRI analyses (Fig. 4), we observed the largest joint hazard ratio for the combination of $PM_{2.5}$, decreased distance to roads and decreased NDVI 300 m (HR = 1.59, 95% CI: 1.40, 1.80) which is larger than the association estimates in the single-exposure models.

4. Discussion

In single-exposure models of the current study, we found that long-term exposure to air pollution, road proximity (living near major roads) was associated with increased risk of PD while the inverse association was found for surrounding green (NDVI). In two-exposure models, the associations with $PM_{2.5}$ and surrounding green persisted while the associations with NO_2 and road proximity generally attenuated to null. The joint HR for the combination of $PM_{2.5}$, road proximity and surrounding green were higher than the HRs from the single-exposure models.

To date, evidence regarding the associations between ambient air pollution and PD was limited (Table S10). For $PM_{2.5}$, as summarized by a recent systematic review and meta-analysis (Kasdagli et al., 2019), studies reported mixed results: positive associations (Kioumourtzoglou et al., 2016; Liu et al., 2016; Shin et al., 2018) and null associations (Cerza et al., 2018; Kirrane et al., 2015; Palacios et al., 2014, 2017; Toro et al., 2019) with the incidence of PD. Moreover, new epidemiological evidence is emerging. A health administrative cohort study incorporating 0.6 million residents in Vancouver, Canada reported a 1.65 μg/m^3 increase in $PM_{2.5}$ was associated with 9% increase in the risk of PD (Yuchi et al., 2020). Another nationwide population-based cohort among 63 million Medicare beneficiaries (aged≥65 years) in the United States from 2000 to 2016 reported the HR was 1.13 (95%CI: 1.12, 1.14) for the first hospital admission of PD following a 5 μg/m^3 increase in long-term $PM_{2.5}$ exposure (Shi et al., 2020). We updated the meta-analysis by adding the findings of these two studies together with the current study, and the updated pooled RR for PD was 1.23 (95% CI:1.02, 1.49) following a 10 μg/m^3 increase in $PM_{2.5}$ exposure although significant heterogeneity still exists (Fig. S2). The newly added three

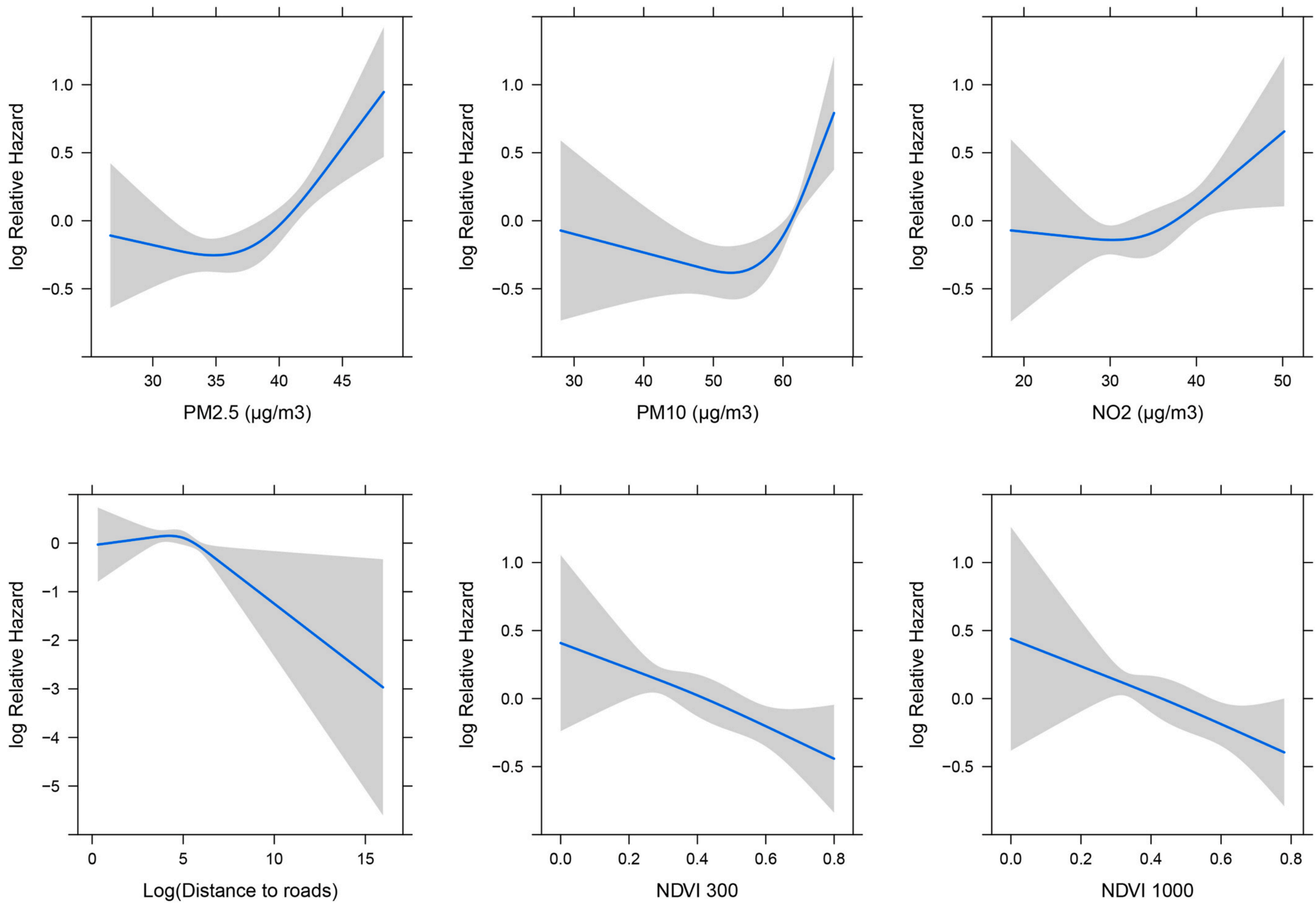

Fig. 2. Concentration-response curves of associations between air pollution, road proximity and surrounding green using restrict cubic splines. All results were adjusted for age, sex, education level, marital status, income, BMI category, smoking, alcohol consumption, tea consumption and physical activity. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

studies had relatively larger effect sizes compared with previous studies and the potential reasons might be: The original $PM_{2.5}$ exposure contrast of the Vancouver study is small (1.65 μg/m^3) and the association estimates were inflated after rescaling to 10 μg/m^3; The US Medicare study was conducted among older adults (≥65 years) with a higher baseline hazard; The current study was conducted among population with a relatively high $PM_{2.5}$ exposure level. Evidence for PM_{10} (Fig. S3) and NO_2 (Fig. S4) was even limited compared with $PM_{2.5}$. More studies are needed to add to the findings.

As most of the previous studies were conducted in developed countries with a relatively low exposure level of $PM_{2.5}$, the current study added to the findings that the adverse effect of $PM_{2.5}$ on the development of PD exists among the Chinese population with a higher $PM_{2.5}$ exposure level. The mean exposure level of $PM_{2.5}$ in present study was higher than previously published studies, which somehow explained the observed higher association estimates. Notably, the exposure-response relationship for $PM_{2.5}$ was flattened when the concentrations were lower than 36 μg/m^3 but increased almost linearly in higher concentrations without evidence of a plateau effect, which underlies the importance of sticking to the current Chinese $PM_{2.5}$ standard (35 μg/m^3 for annual concentrations). Similar exposure-response curves were also observed for PM_{10} and NO_2 (Fig. 2).

For road proximity and PD, only two Canadian studies previously have assessed the association (Chen et al., 2017; Yuchi et al., 2020). A large population-based cohort conducted in Ontario, Canada (Chen et al., 2017) with 31,557 cases found that living near heavy traffic was not associated with risk of PD (HR = 0.99, 95%CI:0.97, 1.01 for log-transformed distance to roads). In contrast, another population-based cohort in Vancouver, Canada with 4201 cases (Yuchi et al., 2020) found that road proximity was positively associated with PD (HR = 1.12, 95%CI:0.91, 1.38 for living within 50 m from highways). The current study found a boardline significant association between road proximity and PD (HR = 0.92, 95%CI: 0.83, 1.02 for log-transformed distance to roads), but the association estimates attenuated when mutually adjusted for air pollution and surrounding green, which is consistent with the Vancouver study, suggesting the observed associations with road proximity may attribute to surrounding green and air pollutants.

Surrounding green was found to be inversely associated with the incident of PD in the current study. The protective effect of greenness was in line with previous findings on other outcomes such as mental health (Klompmaker et al., 2019a), all-cause mortality (Ji et al., 2019) and cardio-metabolic diseases (Chen et al., 2020; de Keijzer et al., 2020; Klompmaker et al., 2019b). Moreover, surrounding green attenuated the associations of air pollutants and road proximity with PD in two-exposure models. The CRI analyses also showed that the joint hazard ratio of exposure to a combination of increased $PM_{2.5}$ and decreased NDVI surrounding green is underestimated by the HRs in the single-exposure models. These results indicated the importance of accounting for potential joint effects of greenness to investigate the association of combined environmental exposure with health-related outcomes.

Plausible mechanism between long-term exposure to air pollution and PD have been raised such as oxidative stress, systemic inflammation

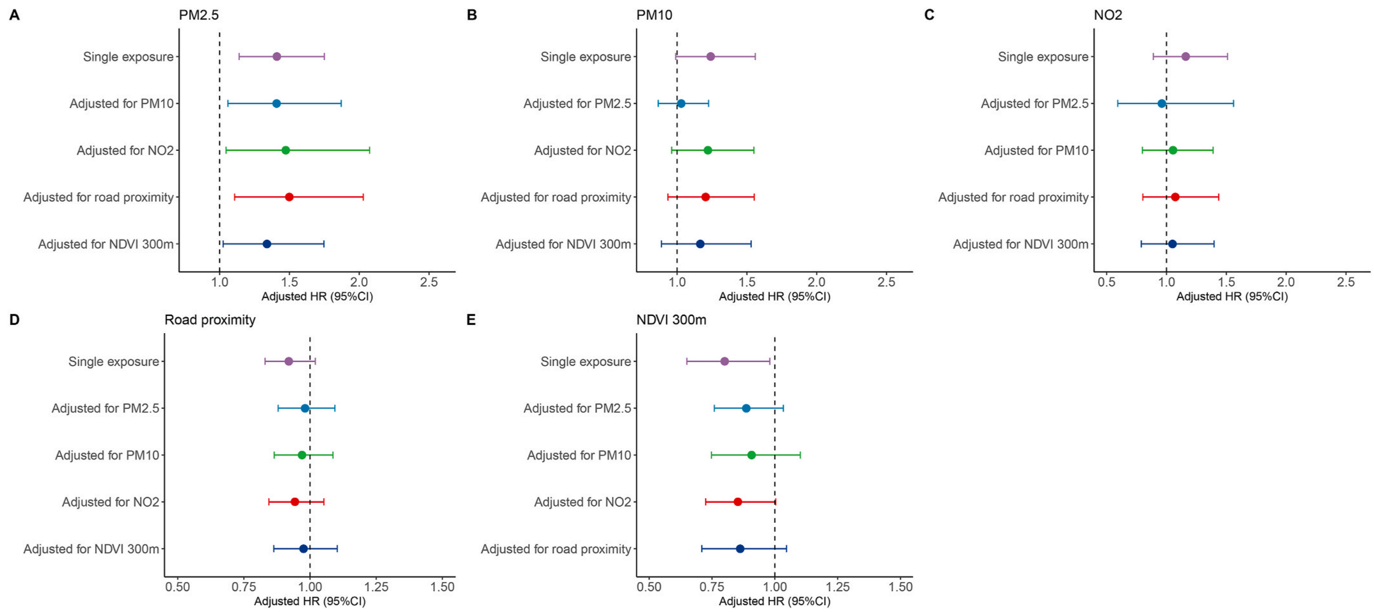

Fig. 3. Associations of air pollution, road proximity, and surrounding green with incidence of Parkinson's disease in single- and two-exposure models. All results were adjusted for age, sex, education level, marital status, income, BMI category, smoking, alcohol consumption, tea consumption and physical activity. All HRs (95% CI) were presented as per interquartile range increment. IQR for $PM_{2.5}$: 8.19 μg/m^3, PM_{10}: 10.79 μg/m^3, NO_2: 11.7 μg/m^3, log-transformed road proximity: 5.95, Normalized Difference Vegetation index (NDVI) 300 m: 0.25. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

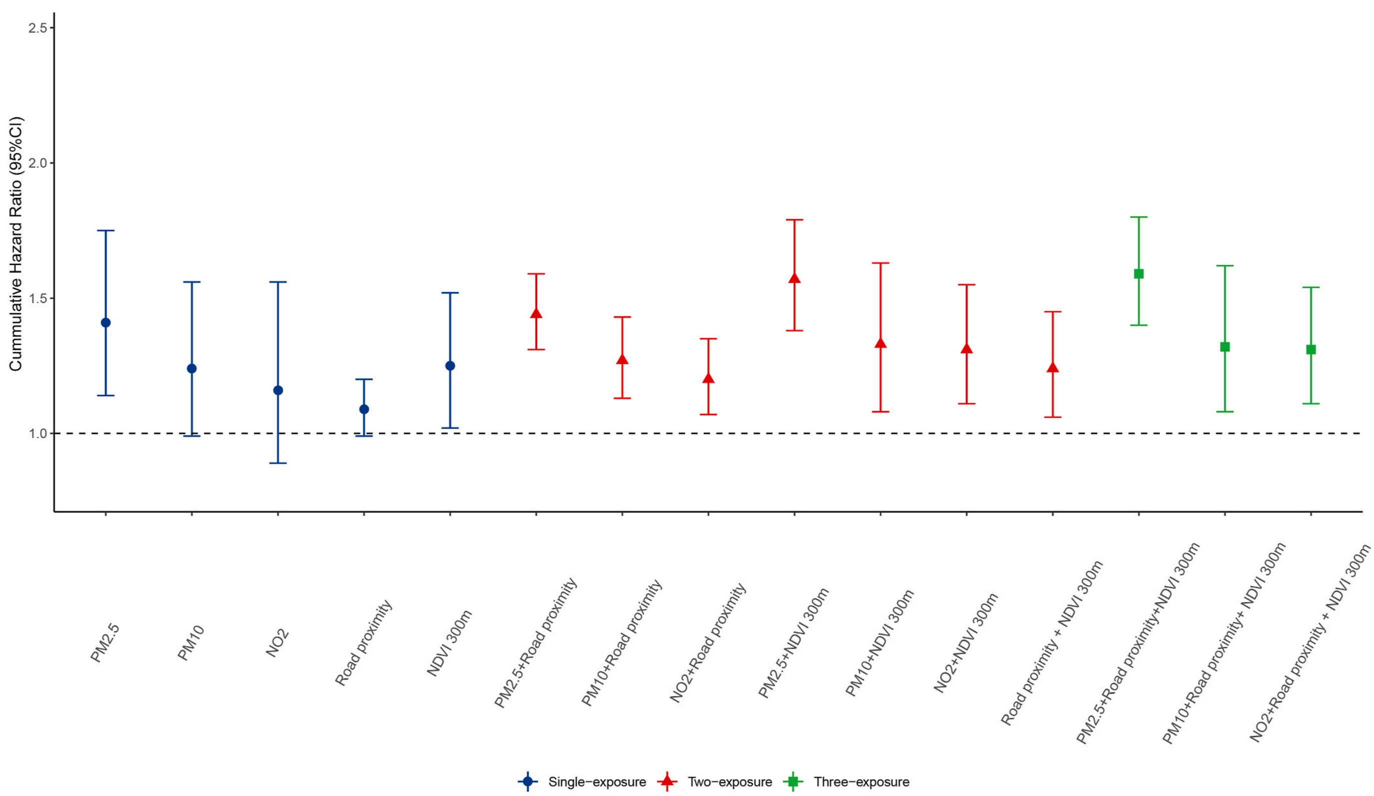

Fig. 4. Joint hazard ratios for associations of air pollution, decreased road proximity and decreased surrounding green with incident of Parkinson's disease. The joint hazard ratios were based on the CRI methods and represents the odds for interquartile range (IQR) increase in air pollution, 1-unit decrease in log-transformed distance to roads and IQR decrease in surrounding green exposure relative to no increase in air pollution (no decrease in road proximity or surrounding green). IQR for $PM_{2.5}$: 8.19 μg/m^3, PM_{10}: 10.79 μg/m^3, NO_2: 11.7 μg/m^3, log-transformed road proximity: 5.95, Normalized Difference Vegetation index (NDVI) 300 m: 0.25. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

and neuroinflammation (Block et al., 2012; Bondy, 2016; Calderon-Garciduenas et al., 2008). Animal experiments showed that exposure to ambient ultrafine particle air pollution would cause a Parkinson's disease phenotype in male mice with locomotor dysfunction and dopaminergic and glutamatergic changes (Allen et al., 2014). Neuropathological analyses of post-mortem tissues from humans living in areas with high air pollution have shown elevations in α synuclein (Calderon-Garciduenas et al., 2008), which has been proposed as a potential indicator for preclinical PD (Chahine and Stern, 2011; Shi et al., 2011). This was confirmed in animal studies that α synuclein in rat brain was elevated after 6-month inhalation exposure to diesel exhaust(Levesque et al., 2011). The beneficial effect of green space has been proposed due to more opportunities for physical activity(Fong et al., 2018; James et al., 2017), better immune function through increased exposure to microorganism associated with plants(Kuo, 2015; Mhuireach et al., 2016), although potential biological pathways between greenness and neurodegeneration remains largely unknown.

The strengths of the current study include the prospective study design, detailed individual-level covariates including education, socioeconomic status and lifestyle factors, and the ability to link to regional HIS to avoid attrition of the cohort, combining the effects of air pollutants, traffic roads, and surrounding greenness. However, there are important limitations that need to be noted when interpreting the current results. First, due to a relatively short follow-up period, the number of incident PD cases was small, which resulted in wide confidence intervals of the association estimates. Second, exposure misclassification may exist as the exposure assessments of air pollution, road proximity and surrounding green were based on residential addresses, which did not account for individual activity patterns (working addresses, exposure during commuting and indoor activities). Third, residential histories may also be a concern. Although we did not have detailed residential histories for participants, sensitivity analyses conducted among non-movers preceding 10 years before the baseline generated essentially similar findings (Table S7). Fourth, noise is another environmental exposure correlated with air pollution, road proximity and surrounding green, which is not considered in the current study due to lack of exposure assessment. This may lead to an underestimation of the joint impact of environmental exposures, although evidence regarding noise is still mixed and scarce on neurologic outcomes (Yuchi et al., 2020). We also acknowledge that the true onset date of PD generally occurred earlier than the diagnosis date, but the exact date of a patient showing symptoms was not available via data linkage to the HIS database. Finally, residual confounding may still exist as information regarding pesticide exposure, head trauma and occupational history was not available in the current study, although the association between these factors and air pollution/surrounding green remains unclear.

5. Conclusions

In this prospective cohort, we found that $PM_{2.5}$, PM_{10} was associated with increased risk of incident PD while surrounding green was associated with decreased risk of PD. The joint impact of exposure to a combination of air pollution and surrounding green may be underestimated by only including one of the correlated exposures (air pollution, surrounding green, road proximity) in single-exposure models. Given the common and pervasive exposure and spatial correlation of these environmental factors, this study suggests that improvements in environmental health policies and land use planning have considerable potential for PD prevention, leading to a broad public health implication.

Credit author contribution statement

Zhebin Yu: Conceptualization, Resources, Writing-review&editing; Fang Wei: Writing-review&editing; Xinhan Zhang: Resources, Validation; Mengyin Wu: Data curation; Hongbo Lin: Resources, Data curation; Liming Shui: Data curation; Mingjuan Jin: Supervision; Jianbing Wang: Supervision; Mengling Tang: Conceptualization, Methodology, Writing-review&editing, Funding acquisition; Kun Chen: Conceptualization, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.envres.2021.111170.

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